

Ch #10

Depreciation

→ Purpose:
To achieve
matching
Principle.

Definition:

- ⇒ Reduction in the cost of Assets after its useful life.
- ⇒ it is an estimated value.
- ⇒ it is the systematic allocation of the depreciable amount of an asset over its useful life.

According to ICMA:

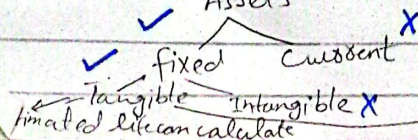
⇒ is diminution of intrinsic value of the asset due to use or laps of time.

⇒ is the systematic allocation of the depreciable amount of an asset over its useful life.

Note:

فرضي 3 fixed Assets Depreciation

Dep Assets



Estimated life cannot be calculate (e.g. land)

Features of Depreciation:

- Non-cash expense
- charged against Profit ↓
- No fixed way of calculation
- Deducted from the value of the Asset in Balance sheet.

Factors in measurement of depreciation:

1. Total cost of Asset / original cost
2. Useful life
3. Residual value / scrap value
4. Maintenance.

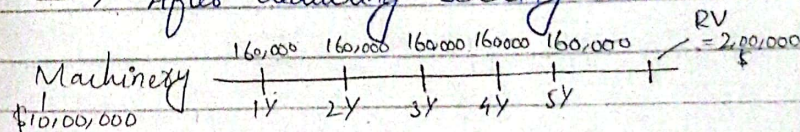
Key Terms

Cost of Assets:

Cost of Assets until put to use. cost of expenses also included like ^{transport} installation cost.

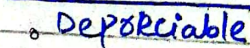
The amount at which sell asset at the end of its useful life is called residual value or scrap value.

→ After deducting selling cost.



• This is the length of time that an asset is considered to be productive.

- Beyond its useful life, it is no longer cost effective to continue.



$$\text{Amount} = \text{Cost} - \text{Residual Value}$$

$$\therefore \text{Depreciation} = \frac{\text{Cost} - \text{Residual value}}{\text{Useful Life}}$$

Methods of Depreciation

1. Straight line method / Fixed installment / cost ^{design}
2. Reducing Balance / Declining balance / Return ^{down value}
_{work down balance}
3. Sum of digit Method
4. MACRS (Modified Accelerated cost
recovery system)

Commercial
life $\rightarrow$ Time Period
during which we take
benefit from them.

Depreciation is charged annually and for complete 12 months except two cases we have to charge dep.

when asset is purchased in 1st year when asset is sold in 1st year

date of purchase date of sale

Straight Line Method

Stoaight Line Method

Also known as fixed installment method.

- Same, depreciation amount

$$\text{Rate} = \frac{\text{Dep}}{\text{Cost}} \times 100 \quad (a)$$

Note:

Note: matters Dates
آگے ISAR اور Asset خرید یا بیچنے کے واسطے whole

2" ¹⁰ month

اگر asset سال کے درمیان خریدے تو ایک سال

بڑھانا شروع ہوا - dep فعالیت کے لیے
Residual value تب آئے گی جب ہم کل depreciate
کریں گے۔

Declining Balance method

(b) Percentage, dep $\frac{1}{2}$ alkaline

• The amount of depreciation will become less and less as the year passes.

- o fixed rate is calculated.

formulae

$$\text{Depreciation of the Year} = \frac{\text{Book value or defining balance}}{\text{Rate of dep}}$$

note

$\frac{1}{2}$ Pec $\rightarrow$ half Yeast
convention
 $\frac{1}{2}$ Yeast
convention

<u>Pros</u>	<u>Cons</u>
◦ No separate calculations are required. ◦ As and become old, depreciation charges become less.	◦ Complex Calculations. ◦ very high rate of depreciation.

Page # 488:-

(a) formula: $\text{Cost} = 325,000$
 $\text{Useful life} = 25 \text{ Years}$
 $\text{Residual value} = 25,000$
 $(\text{Cost} - \text{Residual value})$
 Useful life

Years	Depreciation
9-month 1993	$(325,000 - 25,000) / 5 \times 9/12 = 45,000$
12M 1994	$(325,000 - 25,000) / 5 \times 12/12 = 60,000$
12M 1995	$(325,000 - 25,000) / 5 \times 12/12 = 60,000$
12M 1996	$(325,000 - 25,000) / 5 \times 12/12 = 60,000$
12M 1997	$(325,000 - 25,000) / 5 \times 12/12 = 60,000$
3-month 1998	$(325,000 - 25,000) / 5 \times 3/12 = 15,000$
Totals	300,000

(b)

R.V. = 25,000

formula: $\text{Rate of depreciation} = \frac{1}{\text{Useful life}} \times \text{depreciating balance}$

formula = $\text{Book value} \times \text{rate of dep}$

$$\text{rate of dep} = \frac{1}{5} \times \frac{200}{100}$$

$$= 0.2 \times 2$$

$$= 0.4$$

$$= 40\%$$

Year	Depreciation	Book value
1993	$325,000 \times 0.4 \times 1/2 = 65,000$	$325,000 - 65,000 = 260,000$
1994	$260,000 \times 0.4 = 104,000$	$260,000 - 104,000 = 156,000$
1995	$156,000 \times 0.4 = 62,400$	$156,000 - 62,400 = 93,600$
1996	$93,600 \times 0.4 = 37,440$	$93,600 - 37,440 = 56,160$
1997	$56,160 \times 0.4 = 22,464$	$56,160 - 22,464 = 33,696$
1998	$33,696 \times 0.25 = 8,424$	$33,696 - 8,424 = 25,272$
Total	300,000	

(c)

Sum of digit method

- o Alternate of diminishing balance.
- o Greatest amount is charged in early years.
- o Years in the life of asset added

eg $1+2+3+4+5 = 15$

formula: $\frac{\text{no of years} \times (n+1)}{2}$

$$\text{Depreciation} = \frac{(\text{Cost} - \text{Residual value})}{\text{value at end}}$$

$$\therefore \text{Dep} = \frac{\text{Depreciation}}{\text{Cost}} \times \text{Years fraction}$$

$$\text{Book value} = \text{Dep exp} - \text{Cost}$$

half Year Convention
 Dep value = $325000 - 25000 = 300,000$
 Sum of Years = $\frac{5(5+1)}{2} = \frac{30}{2} = 15$

Years	Depreciation
1993	$300000 \times \frac{5}{15} \times \frac{1}{2} = 50000$
1994	$\frac{30,0000 \times 5/15 \times 1/2}{30,0000 \times 4/15 \times 1/2} = 50000$
1995	$\frac{30,0000 \times 4/15 \times 1/2}{30,0000 \times 3/15 \times 1/2} = 40000$
1996	$\frac{30,0000 \times 3/15 \times 1/2}{30,0000 \times 2/15 \times 1/2} = 30000$
1997	$\frac{300,000 \times 2/15 \times 1/2}{300,000 \times 1/15 \times 1/2} = 20000$
1998	$\frac{300,000 \times 1/15 \times 1/2}{300,000 \times 1/15 \times 1/2} = 10000$
Total	300,000

Note:

اگر half Convention نہ ہو تو 19 1/2
 در دفعہ لیکچر ورکشاپ

$$30,0000 \times \frac{5}{15} =$$

$$300,000 \times \frac{4}{15} =$$

$$300,000 \times \frac{3}{15} =$$

$$300,000 \times \frac{1}{15} =$$

Page # 494

(Exercise 10-6)

Cost = 14000

Residual value = 2000

Useful life = 5 Years

$$\text{Sum of years} = \frac{5(5+1)}{2} = \frac{30}{2} = 15$$

$$\text{Depreciation amount} = 14000 - 2000 = 12000$$

Years	Depreciation
1st	$12000 \times \frac{5}{15} = 4000$
2nd	$12000 \times \frac{4}{15} = 3200$
3rd	$12000 \times \frac{3}{15} = 2400$
4th	$12000 \times \frac{2}{15} = 1600$
5th	$12000 \times \frac{1}{15} = 800$
Total	12000

Page # 494

(EX - 10-2)

Useful life = 5 Years
Cost = 50,000

Salvage value = 5,000

Annual dep of first two Years = ?

(a) Straight line method

Year	Depreciation
1st	$(50,000 - 5,000) \times 1/5 = 9,000$
2nd	$(50,000 - 5,000) \times 1/5 = 9,000$
Total	18,000

(b) Sum of digits method

Dep Cost = $50,000 - 5,000 = 45,000$

Sum of Years = $6(5+1)/2 = 30/2 = 15$

Year	Depreciation
1st	$45,000 \times 5/15 = 15,000$
2nd	$45,000 \times 4/15 = 12,000$
Total	27,000

Page # 494 (EX 10-8)

(a) Straight line method

Useful Life = 5 Years

Cost = 600,000

Residual value = 90,000

Year	Depreciation
1st	$(600,000 - 90,000) \times 1/5 \times 2/12 = 17,000$
2nd	$(600,000 - 90,000) \times 1/5 = 102,000$
3rd	$(600,000 - 90,000) \times 1/5 = 102,000$
4th	$(600,000 - 90,000) \times 1/5 = 102,000$
5th	$(600,000 - 90,000) \times 1/5 = 102,000$
6th	$(600,000 - 90,000) \times 1/5 \times 10/12 = 85,000$
Total	510,000

(b) Straight line with half Year convention

Year	Depreciation
1st	$(600,000 - 90,000) \times 1/5 \times 1/2 = 51,000$
2nd	$(600,000 - 90,000) \times 1/5 = 102,000$
3rd	$(600,000 - 90,000) \times 1/5 = 102,000$
4th	$(600,000 - 90,000) \times 1/5 = 102,000$
5th	$(600,000 - 90,000) \times 1/5 = 102,000$
6th	$(600,000 - 90,000) \times 1/5 \times 1/2 = 51,000$
Total	510,000

Q. (Page # 501) (EX - 103-3)

Cost of equipment = 80,000
 Useful life = 4 Years
 (a) R.V = 5000

Date	Depreciation (Cost - R.V) / Years Useful Life
1993	$(75000/4) \times 4/12 = 6250$
1994	$75000/4 = 18750$
1995	$75000/4 = 18750$
1996	$75000/4 = 18750$
1997	$(75000/4) \times 2/12 = 12500$
Total	75,000

(b) Sum of Years / (half Year convention)

$$= 4(4+1)/2 = 10$$

Date	Depreciation
1993	$75,000 \times 4/10 \times 1/2 = 15000$
1994	$75,000 \times 4/10 \times 1/2 = 15000$
	$75,000 \times 3/10 \times 1/2 = 11250$
	26,250
1995	$75000 \times 3/10 \times 1/2 = 11250$
	$75000 \times 2/10 \times 1/2 = 7500$
	18750
1996	$75000 \times 2/10 \times 1/2 = 7500$
	$75000 \times 1/10 \times 1/2 = 3750$
	11250
1997	$75000 \times 1/10 \times 1/2 = 3750$
	75,000

200%, declining balance

Date	Depreciation	Book value
1993	$80,000 \times 0.4 \times 1/2 = 16000$	$80000 - 16000 = 64000$
1994	$64000 \times 0.4 = 25600$	$64000 - 25600 = 38400$
1995	$38400 \times 0.4 = 15240$	$38400 - 15240 = 23160$
1996	$23160 \times 0.4 = 9264$	$23160 - 9264 = 13896$
1997	$13896 - 5000 = 8896$	8896
Total	75,000	

(Past - Papers 2020 - MIT)

(Switching to straight line when maximizing expense)

Machine purchase on 1-sep, beginning with calendar year 2006

Original Cost = 108,000

Residual value = 8,000

Useful life = 5 Years

Note:

⇒ (Assume the half Year convention method).

(a) Straight line method

Years	Depreciation
2006	$(108,000 - 8,000) \times 1/5 \times 1/2 = 10,000$
2007	$(108,000 - 8,000) \times 1/5 = 20,000$
2008	$(108,000 - 8,000) \times 1/5 = 20,000$
2009	$(108,000 - 8,000) \times 1/5 = 20,000$
2010	$(108,000 - 8,000) \times 1/5 = 20,000$
2011	$(108,000 - 8,000) \times 1/5 \times 1/2 = 10,000$
Total	100,000

(b) 200% decline method

Declining rate = $\frac{1}{5} \times 200 = 40\% = 0.4$

Dep amount = $108,000 - 8,000 = 100,000$

Years	Depreciation	B.V
2006	$108,000 \times 0.4 \times 1/2 = 20,000$	108,000
2007	$108,000 \times 0.4 =$	
2008	$108,000 \times 0.4 =$	
2009	$108,000 \times 0.4 =$	
2010		
2011		
Total		

(c) 150% declining method

Declining rate = $1/5 \times 150 = 30\% = 0.3$

Years	Depreciation	Book value
2006		
2007		
2008		
2009		
2010		
2011		
Total		

(Past - Papers 2020 CS)

Cost of equipment = 325,000

R.V = 25,000

Useful life = 5 Years

Purchase on April 1, 2011.

- (a) Straight line method
- (b) 200% declining balance

(Already solve on previous
Pages Q.1).

Past - Papers - 2023

- Also have same like
question.

Q. Define Accumulated depreciation.

- is a contra asset account that track the total amount of depreciation expense that has been recognized for a particular fixed asset since its acquisition.
- a cumulative record of the asset value decline over time.
- This amount increase each year as the asset continues to depreciate.

Purpose:

- finding net book value
- Tax implications etc.

Q. Define net book value?

⇒ By subtracting accumulated depreciation from the original cost of the asset, we can get the asset's net book value.

⇒ This represents the asset's current value on the balance sheet.

⇒ Different depreciation methods used to calculate depreciation expense, affect the accumulated depreciation expense.